

Institute of Actuaries of India

Subject CS1-Actuarial Statistics (Paper B)

September 2021 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

```
X = c(1:10)
```

```
Lambda = 2
```

(1)

i) `dpois(x, Lambda)`

```
>>Outcome:
```

```
2.706706e-01 2.706706e-01 1.804470e-01 9.022352e-02 3.608941e-02 1.202980e-02
3.437087e-03 8.592716e-04 1.909493e-04 3.818985e-05
```

(1.5)

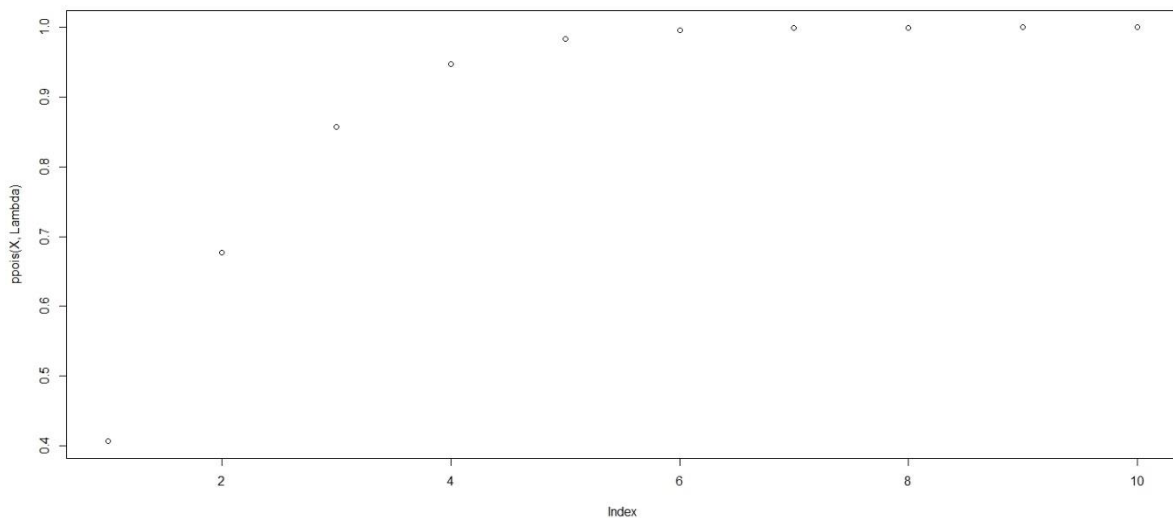
ii) `ppois(x, Lambda)`

```
>> Outcome
```

```
0.4060058 0.6766764 0.8571235 0.9473470 0.9834364 0.9954662 0.9989033 0.9997626
0.9999535 0.9999917
```

(1.5)

iii) `plot(ppois(x,Lambda))`



(1)

[5 Marks]**Solution 2:**

i) a) `Setwd()`

```
Data = read.csv("Temperature_data.csv")
```

```
cor (Data, method = "pearson")
```

	Monday	Tuesday	Wednesday	Thursday	Friday
Monday	1.0000	0.9566	0.5393	0.8236	0.9271
Tuesday	0.9566	1.0000	0.7612	0.9484	0.9776
Wednesday	0.5393	0.7612	1.0000	0.9080	0.7620
Thursday	0.8236	0.9484	0.9080	1.0000	0.9632
Friday	0.9271	0.9776	0.7620	0.9632	1.0000

(3)

b) `cor (Data, method = "kendall")`

	Monday	Tuesday	Wednesday	Thursday	Friday
Monday	1.0000	0.6952	0.2952	0.4667	0.6762
Tuesday	0.6952	1.0000	0.6000	0.7714	0.8286
Wednesday	0.2952	0.6000	1.0000	0.7905	0.5810
Thursday	0.4667	0.7714	0.7905	1.0000	0.7905
Friday	0.6762	0.8286	0.5810	0.7905	1.0000

(3)

c) `cor (Data, method = "spearman")`

	Monday	Tuesday	Wednesday	Thursday	Friday
Monday	1.0000	0.8643	0.4036	0.6250	0.8250
Tuesday	0.8643	1.0000	0.7679	0.9036	0.9536
Wednesday	0.4036	0.7679	1.0000	0.9393	0.7821
Thursday	0.6250	0.9036	0.9393	1.0000	0.8964
Friday	0.8250	0.9536	0.7821	0.8964	1.0000

(3)

- ii) The outcome of Pearson method is based on the values of the data whereas Kendall and Spearman correlation matrix is based on the rank of the values in the dataset. Diagonal in each matrix represents the correlation of temperature with respect to the individual days, for example Monday with Monday. Pearson shows strong positive correlation between temperatures on Monday, Tuesday, Thursday and Friday. Kendall and Spearman shows lesser positive correlation as compared to the Pearson's result (3)
- [12 Marks]**

Solution 3:

- i) If $X_1, X_2, \dots, X_n$ follows Bernoulli distribution with $p = 0.5$ (1)
Then Y follows binomial distribution $\sim \text{Bin}(n, p)$ as sum of independent Bernoulli distribution results in binomial distribution with n trials and p is the probability of success. (1)
- ii) Since 'n' is not specified in the question, sample simulations of Y can be generated assuming any value of 'n'. The answer below is calculated using $n = 10$, however marks are allotted to simulations of Y , calculated using any value of 'n'. [2]

R Code

```
n = 10
p = 0.5
Y ~ rbinom(10, n, p)
Y
```

Values of y from R (since these are random values, answers for each individual may vary from the numbers stated below, please evaluate accordingly)

5 4 4 5 2 6 5 4 2 5 (2)

- iii) The sample simulated in part (ii) are representative value of variable Y itself. Hence, no further calculation required to assess the value of variable Y . (2)
- iv) $\text{dbinom}(4, n, p)$
The solution is based on assumption that $n = 10$ (as assumed in part(ii)). Marks are allotted to solutions calculated using any value for "n".
0.2050781 (1)

[7 Marks]**Solution 4:**

- i) $X = c(0.5820, 0.04981, 0.1552, 0.1555, 0.9036, 0.8501, 0.9288, 0.4408, 0.9688, 0.6300)$
> mean(x)
0.566461 (2)
- ii) Sd(x)

0.3515525

(1)

iii) median(x)
0.606

(3)

[6 Marks]

Solution 5:

i) data=matrix(c(112,38,89,70,130,50,127,75,178,80,210,77,150,68,150,80),4,4)

```
n<-ncol(data)
m<-mean(rowMeans(data))
s<-mean(apply(data,1,var))
v<-var(rowMeans(data))-mean(apply(data,1,var))/n
Z<-n/(n+s/v)
Z*rowMeans(data)+(1-Z)*m
```

[1] 138.08805 64.47793 139.41039 79.02364

Expected total claim payment from

- **insurer A is INR 138.09 Crores**
- **insurer B is INR 64.48 Crores**
- **insurer C is INR 139.41 Crores**
- **insurer D is INR 79.02 Crores**

(1 Mark for correct Z and 1 mark for identifying the correct claim amount for each insurer)

As required for other parts

```
> n
[1] 4
> m
[1] 105.25
> s
[1] 933.8333
> v
[1] 1737.625
> Z
[1] 0.8815584
```

(2)

ii) a) **R Code**

data=matrix(c(112,38,89,70,73,130,50,127,75,87,178,80,210,77,113,150,68,150,80,112),5,4)

#no change in other code from part i

```
n<-ncol(data)
m<-mean(rowMeans(data))
s<-mean(apply(data,1,var))
v<-var(rowMeans(data))-mean(apply(data,1,var))/n
Z<-n/(n+s/v)
Z*rowMeans(data)+(1-Z)*m
```

[1] 137.11712 65.12725 138.41035 79.35279 97.24249

Expected total claim payment from

- **insurer A is INR 137.12 Crores (reduced compared to part i)**
- **insurer B is INR 65.13 Crores (increased compared to part i)**

- insurer C is INR 138.41 Crores (reduced compared to part i)
 - insurer D is INR 79.35 Crores (increased compared to part i)
 - insurer E is INR 97.24 Crores (newly added insurer)
- (1 Mark for correct data and 1 mark for identifying the correct claim amount for each insurer)

required for other parts

> n

[1] 4

> m

[1] 103.45

> s

[1] 824.05

> v

[1] 1288.5

> Z

[1] 0.862154

(2)

b) Comparison of calculated quantities

- With the introduction of insurer E, credibility factor Z has reduced.
- $E[s^2(\text{Theta})]$ (i.e. 's' in the R code used above) has reduced marginally – this has an effect of increasing Z.
- $\text{var}[m(\text{theta})]$ has reduced significantly (i.e. 'v' in the R code used above) - this has an effect of reducing Z.
- Overall the ratio - 's' / 'v' is 0.54 in part i; 's' / 'v' increases to 0.64 in part ii – as value of denominator (in the calculation of $Z = n / (n + (s/v))$) has increased in part ii compared to part i – hence, credibility factor Z has reduced in part ii compared to part i.

(3)

c) Change in expected claim payment for each insurer

Expected claim payment for insurer is based on credibility value and equal to $Z \cdot \text{individual average} + (1-Z) \cdot \text{total average}$

(0.5 mark for comment on formula)

- Total average 'm' has reduced in part ii compared to part i as average of insurer E is lower than total average ($m = 105.25$) in part i

(0.5 mark for comment on 'm')

- Expected claim payment for insurers A and C has reduced in part ii; as lower weight to individual experience compared to part i (A and C have experienced higher claim payments in the past and their average is higher than the total average of 5 insurers 103.45 - value of 'm' as calculated above) brings down the total expected value for these insurers towards 103.45

(1 mark for comment on A and C - where reduction observed)

- Expected claim payment for insurers B and D has increased in part ii; as lower weight to individual experience compared to part i (B and D's individual average values are lower than 103.45) and higher weight to total average (103.45) pulls up the total expected value for these insurers towards 103.45.

(1 mark for comment on B and D – where increase observed)

(3)

iii) a) Use data as used in part ii

```

data=matrix(c(112,38,89,70,73,130,50,127,75,87,178,80,210,77,113,150,68,150,80,112),5,4)
#all other code lines as provide in part iii reference R code
volume<-
matrix(c(165,51,119,108,103,186,63,159,112,116,198,78,219,122,126,200,83,188,133,151),5,4)
n<-ncol(data)
N<-nrow(data)
X<-data/volume #claim payment per unit of volume measure
Xibar<-rowSums(data)/rowSums(volume)
Pi<-rowSums(volume) #volume measure for each insurer
P<-sum(Pi)
Pstar<-sum(Pi*(1-Pi/P))/(N*n-1)
m<-sum(data)/P #average claim payment per unit volume measure across all insurers
s<-mean(rowSums(volume*(X-Xibar)^2)/(n-1))
v<-(sum(rowSums(volume*(X-m)^2))/(n*N-1)-s)/Pstar
zi<-Pi/(Pi+s/v) #credibility factor for each insurer
credibilityi<-zi*Xibar+(1-zi)*m #expected claim payment per unit of volume measure
volume5i<-matrix(c(210,91,192,144,181),5,1) #'volume5i' is Year 5 volume measure
#okay to have 1,5 instead of 5,1 in volume5i (row , column interchange)
claims5i<-credibilityi*volume5i #claims5i is the expected claim payment in year 5

```

Additional output (marks not to be deducted in case if other parts are not provided by student)

```

> n
[1] 4
> N
[1] 5
> X
      [,1] [,2] [,3] [,4]
[1,] 0.6787879 0.6989247 0.8989899 0.7500000
[2,] 0.7450980 0.7936508 1.0256410 0.8192771
[3,] 0.7478992 0.7987421 0.9589041 0.7978723
[4,] 0.6481481 0.6696429 0.6311475 0.6015038
[5,] 0.7087379 0.7500000 0.8968254 0.7417219
> Xibar
[1] 0.7610147 0.8581818 0.8408759 0.6357895 0.7762097
> Pi
[1] 749 275 685 475 496
> P
[1] 2680
> Pstar
[1] 110.0728
> m
[1] 0.7720149
> s
[1] 1.095221
> v
[1] 0.00469698
> zi
[1] 0.7625928 0.5411516 0.7460447 0.6707376 0.6802203
> credibility
[1] 0.7636262 0.8186443 0.8233883 0.6806434 0.7748683
> volume5

```

[,1]
 [1,] 210
 [2,] 91
 [3,] 192
 [4,] 144
 [5,] 181

(2)

b)

> claims5

[,1]
 [1,] 160.36151
 [2,] 74.49663
 [3,] 158.09055
 [4,] 98.01265
 [5,] 140.25116

Claims5 gives expected claim payment based on EBCT 2

- INR 160.36 Crores for insurer A
- INR 74.50 Crores for insurer B
- INR 158.09 Crores for insurer C
- INR 98.01 Crores for insurer D
- INR 140.25 Crores for insurer E

(1 mark for identifying expected claim payment)

Expected claim payments for all insurers has **increased** compared to part ii as it makes use of **Year 5 volume which is higher** for all insurers.

(1 mark for comment on increase and 1 mark for comment on Year 5 volume)

(3)

[15 Marks]**Solution 6:**

- i) H_0 – Company1 and Company2 having equal variance (i.e., ratio of variance =1)
 H_1 – There is a difference in variance between Company1 and Company2.

Company1<-c(1350,1790,1500,1150,2100,2350,1550,1800,1650,1450)

Company2<-c(1500,1200,1300,1700,1800,2400,1450,1950,1850,2100)

var.test(Company1,Company2)

F test to compare two variances

data: Company1 and Company2

F = 0.91388, num df = 9, denom df = 9, p-value = 0.8955

alternative hypothesis: true ratio of variances is not equal to 1

95 percent confidence interval:

0.2269944 3.6792680

sample estimates:

ratio of variances

0.9138781

Conclusion – As the confidence interval for the ratio between variance includes 1, we do not have enough evidence to reject H_0 at 5% significance level. Hence, we can conclude that ratio of variance between Company1 and Company2 is 1 i.e., Company1 and Company2 having equal variance at 5% significance level.

(4)

- ii) H_0 – No difference in mean premium charged by Company1 and Company2.
 H_1 – difference in mean premium charged by Company1 and Company2.

t.test(Company1, Company2, var.equal = TRUE, conf=0.95)

Two Sample t-test

data: Company1 and Company2
 $t = -0.3433$, $df = 18$, $p\text{-value} = 0.7353$
 alternative hypothesis: true difference in means is not equal to 0
 95 percent confidence interval:
 -398.703 286.703
 sample estimates:
 mean of x mean of y
 1669 1725

Conclusion – As the confidence interval for the difference in mean premium includes 0, we do not have enough evidence to reject H_0 at 5% significance level. Hence, we can conclude that there is no difference between mean premium charged by Company1 and Company2 at 5% significance level.

(4)

- iii) H_0 – mean premium charged by Company2 is INR 1500.
 H_1 – mean premium charged by Company2 is greater than 1500.

t.test(Company2, mu=1500, alternative= c("greater"), conf=0.95)

One Sample t-test

data: Company2
 $t = 1.9082$, $df = 9$, $p\text{-value} = 0.04436$
 alternative hypothesis: true mean is greater than 1500
 95 percent confidence interval:
 1508.858 Inf
 sample estimates:
 mean of x
 1725

Conclusion – As the confidence interval for the mean premium does not include 1500, at 5% significance level we have enough evidence to reject null hypothesis.
 (p-value less than 5% also confirms that we have enough evidence to reject null hypothesis at 5% significance level).
 Hence, we can conclude that the mean premium charged by Company2 is greater than INR 1500 at 5% significance level.

(4)

- iv) H_0 – There is no difference in proportion of (policies that result in) claim between Company1 and Company2.

H_1 – There is difference in proportion of (policies that result in) claim between Company1 and Company2.

`prop.test(c(60,50),c(200,100),correct=FALSE)`

2-sample test for equality of proportions without continuity correction

data: c(60, 50) out of c(200, 100)

X-squared = 11.483, df = 1, p-value = 0.0007023

alternative hypothesis: two.sided

95 percent confidence interval:

-0.31677833 -0.08322167

sample estimates:

prop 1 prop 2

0.3 0.5

Conclusion – As the confidence interval does not include 0 (or as p-value is less than 5%), we have sufficient evidence to reject null hypothesis at 5% significance level. Hence, we can conclude that there is difference in proportion of (policies that result in) claims between Company1 and Company2 at 5% significance level.

(4)

v) H_0 – There is no difference in ratio of claim frequency between group1 and other groups in Company2.

H_1 – Claim frequency of group1 is less than other groups in Company2.

`poisson.test(c(65,45),c(250,110), conf=.975, alternative="less")`

Comparison of Poisson rates

data: c(65, 45) time base: c(250, 110)

count1 = 65, expected count1 = 76.389, p-value = 0.01358

alternative hypothesis: true rate ratio is less than 1

97.5 percent confidence interval:

0.0000000 0.9511753

sample estimates:

rate ratio

0.6355556

Conclusion – As the confidence interval for the ratio does not include 1, at 2.5% significance level we have enough evidence to reject null hypothesis.

(p-value less than 2.5% also confirms that we have enough evidence to reject null hypothesis at 2.5% significance level).

Hence, we can conclude that the claim frequency of group1 (having premium less than INR 1500) is less than other groups in Company2.

(4)

[20 Marks]

Solution 7:

i) # Fitting Linear Regression Model

R Code:

`Data_claim = read.csv('HealthClaims.csv')`

`model = lm(CLAIM_AMOUNT~.,data = Data_claim)`

`summary(model)`

R Output:

Call:

```
lm(formula = CLAIM_AMOUNT ~ ., data = Data_claim)
```

Residuals:

```
   Min     1Q  Median     3Q      Max
-155636 -22909  1558  19047 546017
```

Coefficients:

```
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   -32487.4   12119.7  -2.681 0.007435 **
GEOGRAPHYREGION 02  -34130.3    9237.3  -3.695 0.000228 ***
GEOGRAPHYREGION 03  -18468.4   10664.6  -1.732 0.083534 .
GEOGRAPHYREGION 04  -17925.8    9866.1  -1.817 0.069441 .
GEOGRAPHYREGION 05  -18160.8    9895.2  -1.835 0.066668 .
GEOGRAPHYREGION 06    1325.1   10558.3    0.125 0.900146
GEOGRAPHYREGION 07  -30443.9   11716.4  -2.598 0.009463 **
GEOGRAPHYREGION 08  -57724.8    8801.0  -6.559 7.57e-11 ***
GEOGRAPHYREGION 09  -10506.6   11595.9  -0.906 0.365056
GEOGRAPHYREGION 10   30219.2   12332.9    2.450 0.014394 *
GEOGRAPHYREGION 11  -14969.7   11932.6  -1.255 0.209858
PROFESSIONBusiness  158911.2    6700.4  23.717 < 2e-16 ***
PROFESSIONSelfemployed 15605.8    4094.8   3.811 0.000144 ***
PROFESSIONService   163694.3    7081.3  23.116 < 2e-16 ***
GENDERM            -13556.3    3024.6  -4.482 7.99e-06 ***
AGE                2827.4     197.8   14.292 < 2e-16 ***
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 55350 on 1420 degrees of freedom

Multiple R-squared: 0.7275, Adjusted R-squared: 0.7246

F-statistic: 252.8 on 15 and 1420 DF, p-value: < 2.2e-16

R code:

```
anova(model)
```

R Output:

Analysis of Variance Table

Response: CLAIM_AMOUNT

```
      Df   Sum Sq  Mean Sq F value   Pr(>F)
GEOGRAPHY  10 2.2059e+12 2.2059e+11 71.997 < 2.2e-16 ***
PROFESSION   3 8.6687e+12 2.8896e+12 943.124 < 2.2e-16 ***
GENDER       1 1.1570e+11 1.1570e+11  37.763 1.036e-09 ***
AGE          1 6.2585e+11 6.2585e+11 204.271 < 2.2e-16 ***
Residuals 1420 4.3506e+12 3.0638e+09
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

R-Squared : 72.75% of the variation of the claim amount is explained by geographical region, profession, gender and age

Adjusted R- Squared: 72.46% is used to compare with the other models . Adjusted R-Squared is used to compare the goodness of fit for regression models that contain different number of independent variables. The model which maximises the 'adjusted R² statistics can be regarded in some sense as the best model.

p-value of the model is $< 2.2e-16$ which is less than 0.05, hence the null hypothesis that "there is no significant relationship between independent variable and dependent variable" is rejected at 5% level of significance.

P-value of the coefficients: Some of the dependent variables appeared to be insignificant although the overall model is significant. As Geographical region 1, Profession Agriculture and Gender female are insignificant hence clubbed in the intercept itself. It is also observed that Geographical region 10 has a positive coefficient and Geographical region 2 has negative coefficient indicating that claim paid for Geographical region 2 is significantly lower than Geographical region 10. Similarly claim paid for profession business is much higher compared to profession self-employed.

From ANOVA table, it can be concluded that all the variables are significant in terms of claim paid as the p values of all the parameters are less than 0.05 so they are all significantly different from zero

(10)

ii) R Code:
confint(model,level=0.95)

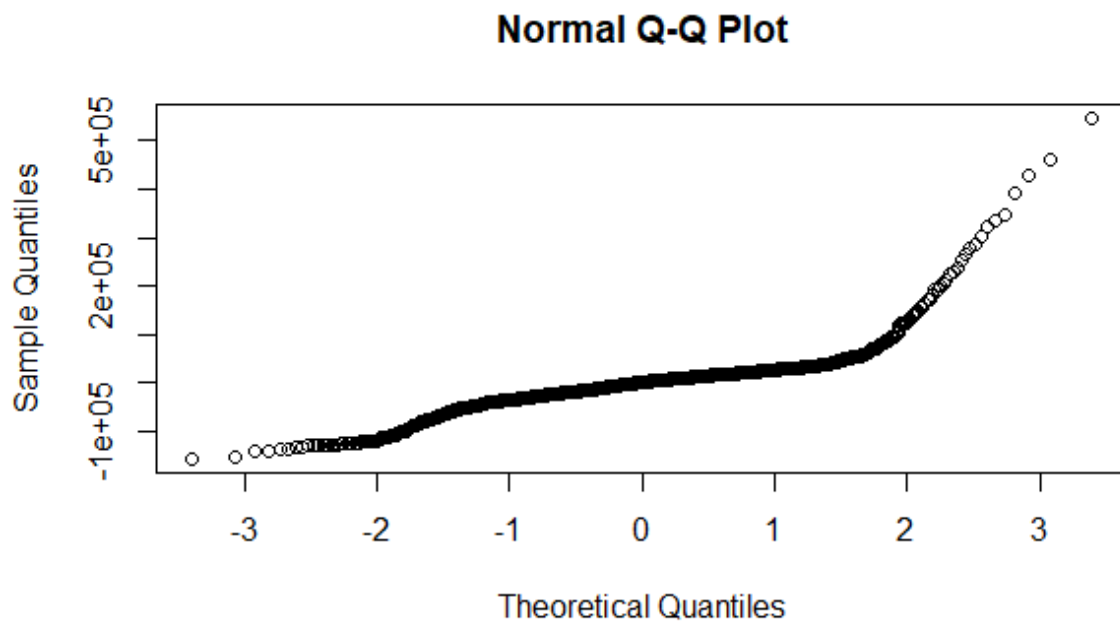
R output:

	2.5 %	97.5 %
(Intercept)	-56261.888	-8712.970
GEOGRAPHYREGION 02	-52250.448	-16010.129
GEOGRAPHYREGION 03	-39388.438	2451.580
GEOGRAPHYREGION 04	-37279.438	1427.815
GEOGRAPHYREGION 05	-37571.441	1249.937
GEOGRAPHYREGION 06	-19386.522	22036.649
GEOGRAPHYREGION 07	-53427.302	-7460.556
GEOGRAPHYREGION 08	-74989.181	-40460.356
GEOGRAPHYREGION 09	-33253.405	12240.270
GEOGRAPHYREGION 10	6026.559	54411.771
GEOGRAPHYREGION 11	-38377.145	8437.686
PROFESSIONBusiness	145767.428	172054.915
PROFESSIONSelfemployed	7573.347	23638.179
PROFESSIONService	149803.318	177585.265
GENDERM	-19489.519	-7623.148
AGE	2439.328	3215.450

(3)

iii) R code:
qqnorm(model\$residuals)

Output:



If the residuals are normally distributed, it is expected the Q-Q plot to be along the diagonal. Here it is not, indicating residuals are not forming normal distribution. Hence linear regression may not be a good fit to the data

(4)

iv) R Code:
`model2 = lm(CLAIM_AMOUNT~.+GEOGRAPHY:PROFESSION + PROFESSION:GENDER + GENDER:GEOGRAPHY,data = Data_claim)`
`summary(model2)`
 Output:
 Call:
`lm(formula = CLAIM_AMOUNT ~ . + GEOGRAPHY:PROFESSION + PROFESSION:GENDER + GENDER:GEOGRAPHY, data = Data_claim)`

Residuals:

Min	1Q	Median	3Q	Max
-194405	-12955	-1034	10489	302012

Coefficients: (3 not defined because of singularities)

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-64594.5	14131.6	-4.571	5.29e-06 ***
GEOGRAPHYREGION 02	-31544.8	13652.9	-2.310	0.02101 *
GEOGRAPHYREGION 03	-28731.6	16539.4	-1.737	0.08258 .
GEOGRAPHYREGION 04	-20891.3	14966.3	-1.396	0.16297
GEOGRAPHYREGION 05	-32366.2	14299.3	-2.263	0.02376 *
GEOGRAPHYREGION 06	-18604.1	15100.7	-1.232	0.21816
GEOGRAPHYREGION 07	-29002.4	15235.1	-1.904	0.05716 .
GEOGRAPHYREGION 08	-33692.9	12994.4	-2.593	0.00962 **
GEOGRAPHYREGION 09	-24352.8	18909.3	-1.288	0.19801
GEOGRAPHYREGION 10	-8309.3	17192.1	-0.483	0.62894
GEOGRAPHYREGION 11	-18947.6	16282.5	-1.164	0.24476
PROFESSIONBusiness	308385.3	15720.6	19.617	< 2e-16 ***
PROFESSIONSelfemployed	30126.2	20435.0	1.474	0.14064

PROFESSIONService	249218.9	42323.2	5.888	4.89e-09	***
GENDERM	-43175.6	14519.3	-2.974	0.00299	**
AGE	3281.3	150.7	21.770	< 2e-16	***
GEOGRAPHYREGION 02:PROFESSIONBusiness	-103364.5	16534.2	-6.252	5.41e-10	***
GEOGRAPHYREGION 03:PROFESSIONBusiness	-61296.8	23584.0	-2.599	0.00945	**
GEOGRAPHYREGION 04:PROFESSIONBusiness	-63068.3	19388.6	-3.253	0.00117	**
GEOGRAPHYREGION 05:PROFESSIONBusiness	-29786.4	18110.5	-1.645	0.10026	
GEOGRAPHYREGION 06:PROFESSIONBusiness	-172656.7	43119.4	-4.004	6.56e-05	***
GEOGRAPHYREGION 07:PROFESSIONBusiness	NA	NA	NA	NA	
GEOGRAPHYREGION 08:PROFESSIONBusiness	-223496.4	15218.8	-14.686	< 2e-16	***
GEOGRAPHYREGION 09:PROFESSIONBusiness	-47277.9	20406.5	-2.317	0.02066	*
GEOGRAPHYREGION 10:PROFESSIONBusiness	135659.4	24962.1	5.435	6.48e-08	***
GEOGRAPHYREGION 11:PROFESSIONBusiness	NA	NA	NA	NA	
GEOGRAPHYREGION 02:PROFESSIONSelfemployed	-14904.8	20843.9	-0.715	0.47469	
GEOGRAPHYREGION 03:PROFESSIONSelfemployed	-5376.4	22863.8	-0.235	0.81413	
GEOGRAPHYREGION 04:PROFESSIONSelfemployed	-1332.0	21623.4	-0.062	0.95089	
GEOGRAPHYREGION 05:PROFESSIONSelfemployed	-715.2	21573.5	-0.033	0.97356	
GEOGRAPHYREGION 06:PROFESSIONSelfemployed	2571.7	23542.6	0.109	0.91303	
GEOGRAPHYREGION 07:PROFESSIONSelfemployed	2053.5	23634.2	0.087	0.93077	
GEOGRAPHYREGION 08:PROFESSIONSelfemployed	-21048.6	20323.1	-1.036	0.30053	
GEOGRAPHYREGION 09:PROFESSIONSelfemployed	3725.3	25207.1	0.148	0.88253	
GEOGRAPHYREGION 10:PROFESSIONSelfemployed	-1907.5	25341.3	-0.075	0.94001	
GEOGRAPHYREGION 11:PROFESSIONSelfemployed	-10631.8	26202.1	-0.406	0.68498	
GEOGRAPHYREGION 02:PROFESSIONService	-91701.8	44213.1	-2.074	0.03826	*
GEOGRAPHYREGION 03:PROFESSIONService	-32757.7	46667.3	-0.702	0.48283	
GEOGRAPHYREGION 04:PROFESSIONService	108004.4	47768.4	2.261	0.02391	*
GEOGRAPHYREGION 05:PROFESSIONService	-39108.5	48885.0	-0.800	0.42384	
GEOGRAPHYREGION 06:PROFESSIONService	93708.9	44227.6	2.119	0.03429	*
GEOGRAPHYREGION 07:PROFESSIONService	NA	NA	NA	NA	
GEOGRAPHYREGION 08:PROFESSIONService	-174071.2	42916.8	-4.056	5.27e-05	***
GEOGRAPHYREGION 09:PROFESSIONService	-24666.4	46726.3	-0.528	0.59766	
GEOGRAPHYREGION 10:PROFESSIONService	108119.5	47621.0	2.270	0.02334	*
GEOGRAPHYREGION 11:PROFESSIONService	30088.2	46092.1	0.653	0.51400	
PROFESSIONBusiness:GENDERM	-69347.9	8414.3	-8.242	3.91e-16	***
PROFESSIONSelfemployed:GENDERM	-11465.1	5210.5	-2.200	0.02794	*
PROFESSIONService:GENDERM	-35812.2	8955.2	-3.999	6.70e-05	***
GEOGRAPHYREGION 02:GENDERM	48893.3	15503.6	3.154	0.00165	**
GEOGRAPHYREGION 03:GENDERM	47556.3	17387.1	2.735	0.00632	**
GEOGRAPHYREGION 04:GENDERM	29240.4	16539.5	1.768	0.07730	.
GEOGRAPHYREGION 05:GENDERM	47373.9	16304.1	2.906	0.00372	**
GEOGRAPHYREGION 06:GENDERM	31561.7	17303.9	1.824	0.06837	.
GEOGRAPHYREGION 07:GENDERM	41765.9	18709.7	2.232	0.02575	*
GEOGRAPHYREGION 08:GENDERM	47600.7	14846.0	3.206	0.00138	**
GEOGRAPHYREGION 09:GENDERM	36413.1	20344.7	1.790	0.07370	.
GEOGRAPHYREGION 10:GENDERM	9577.3	19749.4	0.485	0.62779	
GEOGRAPHYREGION 11:GENDERM	31022.5	19076.4	1.626	0.10413	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 40270 on 1380 degrees of freedom

Multiple R-squared: 0.8598, Adjusted R-squared: 0.8542

F-statistic: 153.9 on 55 and 1380 DF, p-value: < 2.2e-16

```
anova(model3)
```

Analysis of Variance Table

Response: CLAIM_AMOUNT

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
GEOGRAPHY	10	2.2059e+12	2.2059e+11	136.0055	< 2.2e-16 ***
PROFESSION	3	8.6687e+12	2.8896e+12	1781.5995	< 2.2e-16 ***
GENDER	1	1.1570e+11	1.1570e+11	71.3357	< 2.2e-16 ***
AGE	1	6.2585e+11	6.2585e+11	385.8770	< 2.2e-16 ***
GEOGRAPHY:PROFESSION	27	1.9432e+12	7.1969e+10	44.3731	< 2.2e-16 ***
PROFESSION:GENDER	3	1.2973e+11	4.3244e+10	26.6624	< 2.2e-16 ***
GEOGRAPHY:GENDER	10	3.9538e+10	3.9538e+09	2.4378	0.007001 **
Residuals	1380	2.2382e+12	1.6219e+09		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Comments:

- R-Squared and Adjusted R-Squared improved and hence model is a better fit compared to the initial model
- Interaction effect between geographical region and profession, profession and gender, gender and geographical region along with their main effects emerged out to be significant as demonstrated by ANOVA table

(10)

v) R code:

```
model3 = lm(log(CLAIM_AMOUNT)~.,data = Data_claim)
summary(model3)
```

Output:

Call:

```
lm(formula = log(CLAIM_AMOUNT) ~ ., data = Data_claim)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.86313	-0.18493	-0.00393	0.17478	1.02901

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	8.4529628	0.0580014	145.737	< 2e-16 ***
GEOGRAPHYREGION 02	-0.1408025	0.0442069	-3.185	0.00148 **
GEOGRAPHYREGION 03	-0.0774399	0.0510375	-1.517	0.12941
GEOGRAPHYREGION 04	-0.0889472	0.0472161	-1.884	0.05979 .
GEOGRAPHYREGION 05	-0.0829154	0.0473553	-1.751	0.08018 .
GEOGRAPHYREGION 06	0.0231982	0.0505290	0.459	0.64623
GEOGRAPHYREGION 07	-0.0654231	0.0560714	-1.167	0.24349
GEOGRAPHYREGION 08	-0.3693971	0.0421191	-8.770	< 2e-16 ***
GEOGRAPHYREGION 09	-0.0568913	0.0554943	-1.025	0.30546
GEOGRAPHYREGION 10	0.0540411	0.0590215	0.916	0.36002
GEOGRAPHYREGION 11	-0.0345033	0.0571059	-0.604	0.54581
PROFESSIONBusiness	0.8217253	0.0320661	25.626	< 2e-16 ***
PROFESSIONSelfemployed	0.1956622	0.0195963	9.985	< 2e-16 ***

```

PROFESSIONService    0.8736364 0.0338891 25.779 < 2e-16 ***
GENDERM              -0.0889058 0.0144749 -6.142 1.06e-09 ***
AGE                  0.0536827 0.0009467 56.703 < 2e-16 ***

```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2649 on 1420 degrees of freedom

Multiple R-squared: 0.9227, Adjusted R-squared: 0.9219

F-statistic: 1130 on 15 and 1420 DF, p-value: < 2.2e-16

R code:

```
anova(model3)
```

Analysis of Variance Table

Response: log(CLAIM_AMOUNT)

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
GEOGRAPHY	10	246.08	24.608	350.69	< 2.2e-16 ***
PROFESSION	3	706.84	235.613	3357.70	< 2.2e-16 ***
GENDER	1	11.25	11.255	160.39	< 2.2e-16 ***
AGE	1	225.62	225.616	3215.24	< 2.2e-16 ***
Residuals	1420	99.64	0.070		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Key differences:

- R-Squared and Adjusted R-Squared increased to above 90% and hence model is a better fit compared to the earlier models
- The significance level of few factor coefficients changed when compared with the model in (i) however ANOVA table shows there is no change in number of significant variables

(8)

[35 Marks]
